COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	09/30/2026	**DATE SUBMITTED:**	09/30/2026
NAME:	Jan Mikaela Guarin Jiro Luis Katindig Juan Paulo Lara Gian Carlo G. Libo-on Ramelissa Melith V. Limpio	**INSTRUCTOR:**	Engr. Jimlord Quejado

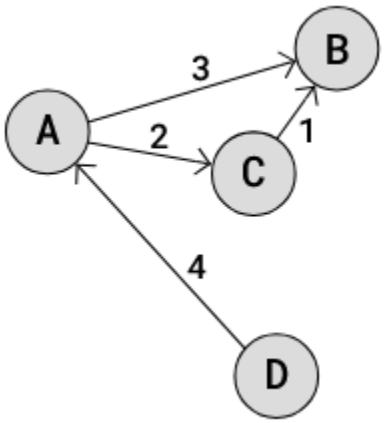
SEATWORK NO. 10.1 GRAPHS

1. PROCEDURE OUTPUT

What procedures did you perform?

1. Define the following: (10)
   - A. Graph
   - A graph is a non-linear data structure composed of fundamental points called vertices (or nodes) that are linked together by edges.
   - B. Vertex
   - A vertex represents a distinct entity in a graph, identified by a unique key, and can store supplementary data known as a payload.
   - C. Nodes
   - The fundamental elements or points in a graph that are connected to one another through edges.
   - D. Weight
   - A value associated with an edge representing the cost or distance of traversing from one vertex to another.
   - E. Path
   - A sequence of connected vertices. Its length is measured by the number of edges for unweighted paths or the sum of edge weights for weighted paths.
   - F. Directed graph
   - A graph where all edges are one-way ordered pairs (v,w), permitting movement from v to w but not necessarily in reverse.
   - G. Connected graph
   - A graph where a path exists between every pair of vertices, making every vertex reachable from any other. Otherwise, it is disconnected.
   - H. Directed cyclic
   - A directed graph containing at least one cycle, which is a path that starts and ends at the same vertex, a directed graph with no cycles is called a DAG (Directed Acyclic Graph).
   - I. Adjacency matrix
   - A 2D matrix implementation where rows and columns represent vertices and cells store edge connections or weights, consumes v^2 memory and wastes space on sparse graphs.
   - J. Adjacency List
   - A space-efficient graph implementation using a master list where each vertex maintains a map of its adjacent neighbors and edge weights.

2. Identify the parts of the following graph. (10)



Vertices/Nodes:
- A, B, C, D

Edges:
- A -> B
- A -> C
- C -> B
- D -> A

Weights:
- A -> B = 3
- A -> C = 2
- C -> B = 1
- D -> A = 4

Paths:
- A -> B
- A -> C -> B
- D -> A -> B
- D -> A -> C -> B

Type of Graph:
- Directed weighted graph

3. Plot the adjacency matrix of the graph above. (10 pts)

	A	B	C	D
A	0	3	2	0
B	0	0	0	0
C	0	1	0	0
D	4	0	0	0

4. Create a program in C++ to create a graph that looks like in the above figure.

New.cpp > main()

```cpp
#include <iostream>

int main() {
    char nodes[] = {'A', 'B', 'C', 'D'};

    // 4x4 Adjacency Matrix
    int adjMtrx[4][4] = {
        {0, 3, 2, 0},
        {0, 0, 0, 0},
        {0, 1, 0, 0},
        {4, 0, 0, 0}
    };

    // Printing the matrix
    std::cout << "  ";
    for (int i = 0; i < 4; i++) {
        std::cout << " " << nodes[i];
    }
    std::cout << "\n";
    for (int i = 0; i < 4; i++) {
        std::cout << nodes[i] << " ";
        for (int j = 0; j < 4; j++) {
            std::cout << " " << adjMtrx[i][j];
        }
        std::cout << "\n";
    }

    return 0;
}
```

Figure 1. Code snippet

```
  A B C D
A 0 3 2 0
B 0 0 0 0
C 0 1 0 0
D 4 0 0 0
```

Figure 2. Output

5. Differentiate BFS vs DFS types of search.

Type	Traversal	Data Structure	Path (unweighted graph)	Target	Memory Allocation
BFS	Level by Level	Uses QUEUE (FIFO)	Finds the Shortest Path in an unweighted graph	Close to the starting position	Higher for wider graphs
DFS	Sub-tree by Sub-tree	Uses STACK (LIFO)	Not guaranteed to find the shortest path	Far from the starting position	Lower for deeper graphs

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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